

Case 3 Pot of Gold? The US Legal Marijuana Industry*

The inauguration of Joe Biden as President of the United States of America on January 22, 2021 was viewed as a watershed event for the marijuana industry. Although legalization of marijuana production and distribution had spread to a majority of US states, continuing illegality of marijuana at the federal level had constrained the industry's development. On the campaign trail, Biden had spoken in favor of decriminalizing marijuana possession, but had given no indication that marijuana would be removed from the federal government's list of illegal drugs.

Despite a confused legal situation, the marijuana industry was riding a wave of optimism. Sales of legal marijuana were estimated at \$18.5 billion in 2020, and US marijuana companies had raised some \$120 billion in finance during that year. During the early months of 2021, projections of continued growth in marijuana consumption together with an upsurge in investment flows—both venture capital and initial public offerings (IPOs)—had generated a buzz of excitement. Some of the leading investors are shown in Table 1. The aggregate market capitalization of the 28 publicly listed marijuana companies in the US Marijuana Stock Index was \$65 billion at the end of March 2021. Management consulting firm, A.T. Kearney pointed to “a potential multibillion-dollar opportunity, anchored by a portfolio of cannabis-based products ranging from beverages and OTC drugs to cosmetics, skincare items, and beyond” and observed that “cannabis legalization has created a 21st-century gold rush.”¹

However, beyond the buzz of excitement and the influx of entrepreneurs and money that the industry had attracted, there remained perplexing questions over marijuana's potential to generate attractive profits. Would the industry offer the sustained high profitability of the two other heavily regulated industries supplying recreational drugs—alcohol and tobacco—or would it suffer the squeezed margins and low returns typical of the agricultural sector?

Legalization

In 1996, California became the first state to legalize the sale of marijuana for medical use. Then, in 2014, Colorado and Washington became the first states to legalize the production, sale, and consumption of marijuana for recreational use. At the beginning of 2021, the use of marijuana for medical purposes, if recommended by a doctor, was legal in 35 states, while 15 states had legalized marijuana for recreational purposes.²

*This case was prepared by Robert M. Grant. © 2021 Robert M. Grant.

TABLE 1 Private equity and venture capital funds investing in marijuana businesses

Fund	Founded	Size	Notes
Privateer Holdings	2010	\$228m.	Backed by Peter Thiel's Founders Fund. Investments include Tilray (medical marijuana, The Goodship, (marijuana-infused candies and cookies), Marley Natural, and Leafly
Tuatara Capital	2014	\$93m.	Investments include Willie's Reserve (owned by Willie Nelson), TeeWinot Life Sciences (cannabis-based pharmaceuticals), Green Dot Labs (cannabis extracts)
MedMen Capital	2016	\$383m.	Parent company, MedMen Enterprises, or MedMen, is a publicly traded, US-based cannabis company with operations in California, Nevada, Arizona, Florida, Illinois, New York, and Massachusetts. It owns 29 retail stores and 6 cultivation facilities. Listed on Canadian Securities Exchange with valuation of \$95m.
Poseidon Asset Management	2013	\$145m.	Invests in agricultural technology and data analytics for the marijuana industry
Salveo Capital	2017	\$121m.	Investments include Headset (cannabis data analytics) and Front Range Biosciences (biotech)
Casa Verde	2015	\$45m.	Invests mainly in support services for cannabis industry including Eaze Solutions Inc. (medical marijuana home delivery service) and Trelis, Green Bits, and FunkSac (marijuana packaging)
Phyto Partners	2015	\$14m.	Invests in digital technologies for cannabis industry including New Frontier (data analytics), Grownetics (growing solutions), and Flowhub (business management platform)

Source: Based on Mrinalini Krishna, Top Marijuana Private Equity and Venture Capital Funds, Nov 12, 2020. <https://www.investopedia.com/news/top-marijuana-private-equity-and-venture-capital-funds/>.

Yet, continuing illegality of the production, sale, and possession of marijuana under federal law was a major handicap for the industry. In particular, firms engaged in producing and selling marijuana had limited access to the US financial system. Banks were fearful that involvement with the industry might contravene drug-racketeering or money-laundering rules. In addition, it was not possible for marijuana producers or distributors to register patents or trademarks with the US Patent Office.

Among the 35 states that had not legalized recreational use of marijuana, 15 had decriminalized its possession, at least for first offenses. However, laws against marijuana consumers continued to be enforced. In 2019, there were 545,602 arrests of people on marijuana-related charges. Over 90% of these arrests were for possession, and the arrest total had declined only modestly since 2013. At the federal level, even in the absence of legislative change, it seemed unlikely that the Justice Department would seek to counteract legalization at the state level.

The Market for Marijuana

The US market for marijuana can be segmented between legal and illegal sectors and between medical and recreational use.

TABLE 2 Estimates of the US market for marijuana

Market feature	Data
Numbers of users, 2017	Total users: 33m. (Gallup poll); 55m. (Marist poll) Medical marijuana users: 1.7m. (Marijuana Business Daily)
Marijuana sales, 2017	Total legal sales: \$9bn. (BDS Analytics), \$5.1–6.1bn. (Marijuana Business Daily) Medical sales: 60% of total; recreational 40% of total Total illegal sales: \$21bn. (New Frontier), \$64.7bn. (ArcView)
Leading states for legal marijuana sales, 2017	California \$1.45bn.; Washington \$7.8bn.; Colorado \$0.44bn.; Arizona \$0.35bn.; Michigan \$0.13bn.; Illinois \$0.08bn.; Oregon \$0.07bn. (Marijuana Business Daily)
Number of legal marijuana businesses, 2017	Growers: 2500–3500; retail dispensaries 3300–4300; infused product manufacturers: 1600–2000; testing labs: 100–150 (Marijuana Business Daily)

Note: The sources of the estimates are shown in brackets.

Recreational consumption of marijuana in the United States has been widespread since the mid-1960s, although estimates of its prevalence are imprecise. Table 2 shows the estimated size of the US market; Table 3 shows estimated sales by state. The United States appears to have one of the highest rates of marijuana consumption among the advanced industrialized nations. Statista estimates that 17.5% of all Americans consumed marijuana in 2019—up from 11% in 2000. Marijuana use is most prevalent among adolescents and young adults: in 2018, 35.7% of 12th graders had used marijuana during the year prior to the survey and 22.3% in the past month.³ However, the biggest increase in marijuana use in recent years has been among older Americans: the proportion of over-65s using marijuana increased from 2.4% in 2015 to 4.2% in 2018.⁴ Future consumption trends for recreational marijuana depended upon the extent to which growing legality and social acceptability encouraged marijuana consumption within mainstream society.

Medical use of marijuana has grown with increasing recognition of its health benefits. These health benefits are associated with both of marijuana's active ingredients: CBD and THC. CBD (also known as cannabidiol) is nonpsychoactive, although not approved by the FDA. CBD is sold over-the-counter as an ingredient in oils, oral sprays, creams, pills, and edibles (such as gummies and beverages) in most states. Its claimed benefits include relief from pain, anxiety, and depression. It has demonstrated particular efficacy in helping people living with cancer. In 2018, the FDA gave approval to Epidiolex, the first prescription medication to contain CBD and used to treat epilepsy.

THC offers many of the same medical benefits as CBD—notably the relief of pain, nausea, and inflammation. Typically, marijuana grown for medical use has higher CBD and lower THC content. Most CBD is produced from hemp-marijuana with low THC content—which can be grown freely by farmers without any need for licensing. As Table 4 shows, California is the major market for medical marijuana.

Figure 1 shows projections of marijuana sales to both recreational and medical markets.

TABLE 3 Sales of marijuana (legal and illicit) by state, 2018 (\$ millions)

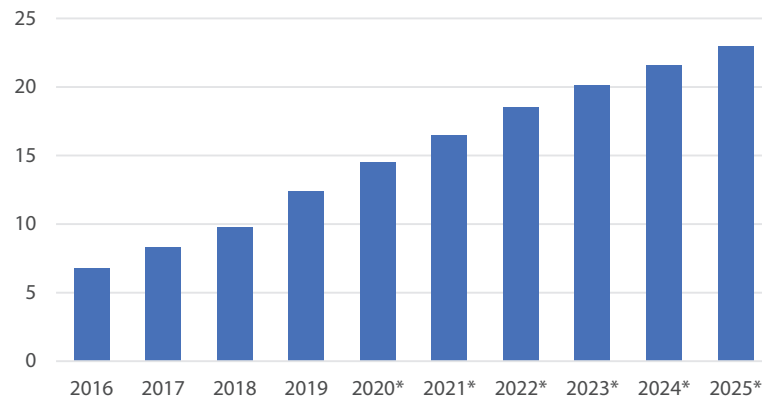
1	California	5600
2	New York	2300
3	Washington	2200
4	Florida	2000
5	Colorado	1900
6	Texas	1700
7	Michigan	1500
8	Massachusetts	1100
9	Oregon	1000
10	Arizona	970
11	Georgia	884
12	North Carolina	875
13	Nevada	671
14	Wisconsin	489
	Other states	3185
	Total	26,374

Source: New Frontier Data.

TABLE 4 States with the most registered medical marijuana patients, 2020

1	California	1,920,294
2	Florida	306,185
3	Michigan	261,989
4	Arizona	230,317
5	New York	114,947
6	Illinois	113,458
7	Maryland	99,912
8	Ohio	94,356
9	New Mexico	82,147
10	Colorado	81,722
11	Massachusetts	69,000
12	Maine	65,368
13	Arkansas	50,377
14	Washington	44,532
15	Missouri	44,097

Source: Marijuana Policy Project.

FIGURE 1 Sales of legal marijuana in the United States (\$ billions)

Source: Sales of legal cannabis in the United States from 2016 to 2025. Statista.

Development of the Industry

The dominant influence on the development of the legal marijuana industry has been the evolving legal and regulatory framework for marijuana. Legalization of the production, sale, and possession of marijuana has been accompanied by the development of complex regulatory structures that differ greatly from state to state.

The main entry barrier into the industry is the need for every marijuana business to obtain a license. The cost and availability of licenses vary by state (see Table 5). Colorado has strict eligibility criteria: licensees must be US citizens, be state residents, have clean criminal records, and meet other standards, but there is no quantitative limit on licenses issued. In Washington, a fixed number of licenses are available, and their allocation is by lottery. In California, eligibility criteria are relaxed with no limit to the number of licenses that can be issued. In Illinois, Massachusetts, and Arizona, there are limits to the number of licenses that the state can issue.

However, in all states, opening a marijuana cultivating or dispensing business requires local authorization as well as a state license. In Colorado, Washington, and California, many counties and cities decided not to permit marijuana businesses. As a result, most businesses are concentrated in relatively few locations. In Colorado, there are 1021 medical and recreational dispensaries, of which 446 are in the Denver metro area. A large proportion of California's dispensaries are in five cities: San Francisco, Oakland, San Jose, Sacramento, and Los Angeles.

In addition to the conditions for obtaining a license, the industry is subject to an array of regulatory requirements. All marijuana facilities must have elaborate security equipment installed, including surveillance cameras and precautions against theft. Each marijuana plant must be tracked—this typically requires RFID tagging. The physical movement of marijuana is further subject to regulations that relate to specifications for the vehicles that can be used to transport marijuana and the individuals authorized to undertake transportation.

The issuing of separate licenses for different types of marijuana businesses—growers, distributors/transporters, manufacturers, retailers, and testing labs—tends to reinforce fragmentation along the industry's value chain. In some states, dispensaries are required to supply their own marijuana, therefore ensuring vertical integration in those jurisdictions.

TABLE 5 Marijuana licensing in selected states, 2019

	Production	Retail	Licensing fees
Colorado	Recreational: 716 Medical: 662	Recreational: 553 Medical: 468	\$2500–\$22,000
Washington	982	Recreational: 556 Medical: 117	Application fee: \$250 Annual licensing fee: \$1480
California	>3200	1440	Licensing administered at county and city level
Oregon	Recreational: 1115 Medical: 4910	Recreational: 606 Medical: n.a.	Dispensary: \$4000 or \$4750 Production: \$1000–\$5750
Illinois	22	54	Production: \$22,000 Retail: \$5000
Massachusetts	Dispensaries grow their own	Recreational: 8 Medical: 12	Application fees: \$2000–\$30,000 Licensing fees: \$625–\$50,000
Nevada	64	47	Application fee: \$5000 Licensing fees: \$10,000–\$30,000
Arizona	Dispensaries grow their own	131	\$5000

Most states encourage small businesses—including social enterprises—in developing their marijuana industries. However, the flood of finance into the industry and the desire to exploit economies of scale have encouraged a wave of consolidation through mergers and acquisitions together with a race for organic growth among leading players. In Colorado, larger chains include Medicine Man (26 stores), The Green Solution (21 stores), and Native Roots (20 stores).

Several of the larger players operate both medical and recreational facilities, are vertically integrated from production to retail, and have operations in multiple states. For example:

- Curaleaf Holdings is based in Massachusetts and owns dispensaries and growing facilities in 18 states. In July 2020, it acquired the Chicago-based chain, Grassroots, for \$830 million. Its revenues during 2020 exceeded \$1 billion.
- Trulieve is based in Florida where it operates over 40 dispensaries. It also has dispensaries in Connecticut, California, and Massachusetts.
- Harvest Health and Recreation, Inc. operates 37 retail locations in seven states, mainly Arizona, Florida, Maryland, and Pennsylvania. It had revenues of \$350m. in 2020.
- Acreage Holdings, Inc. operates 71 dispensaries and 24 cultivation and processing facilities in 15 states. Its 2020 revenues topped \$150m.
- MedMen Enterprises has 29 retail stores and 6 cultivation facilities across California, Nevada, Arizona, Florida, Illinois, New York, and Massachusetts. It earned \$164m. in revenues during 2020.

The development of the industry has been accompanied by the development of an infrastructure of support services:

- Helix TCS and MJ Freeway are Denver-based suppliers of “seed-to-sale” tracking and compliance software.
- Advanced Cannabis Solutions leases real estate to growers and dispensaries.
- Nexus Corp. and GrowGeneration Corp. supply greenhouses and hydroponic growing systems to marijuana growers.
- KushCo Holdings provides packaging.
- ArcView Group provides a hub for data, research, media, and consulting. Its ArcView Investor Network provides a forum linking accredited investors to cannabis businesses.

The Economics of the Marijuana Business

Cultivation

Growing marijuana, whether for the medical or the recreational market, requires first a license, and then the acquisition of a growing facility. Most marijuana is grown indoors in climate-controlled warehouses under artificial light. Cultivation in greenhouses and outdoors offers economies both in setup and energy costs; these advantages are mitigated by the need for extensive security equipment for all marijuana-growing facilities. Indoor cultivation also permits multiple growing cycles each year. The average size of marijuana-growing facilities grew from 25,300 square feet in 2017 to 36,300 in 2020.⁵ However, largest facilities are over 250,000 square feet and America’s biggest marijuana greenhouse is Copperstate Farms 1.7 million square feet facility in Arizona.

The cultivation cycle involves the following stages:

- 1 Establishing stage: cloning new plants from existing female plants and allowing the new plants 7–12 days to become established.
- 2 “Veg” (or growing) stage: two months under constant light.
- 3 Flowering stage: about two months with a daily cycle of 12 hours of light followed by 12 hours of darkness.
- 4 Processing stage: hanging the plants upside down, then harvesting their flowers (“buds”).
- 5 Curing stage: drying the flowers.

Estimates of revenues and costs of marijuana growing suggest that it is a highly profitable crop. The median costs of producing a pound of marijuana in 2020 were \$100 outdoors, \$233 in greenhouses, and \$396 indoors (these costs had fallen by about 8% from 2018).⁶ Wholesale prices per pound varied substantially by location and quality. Outdoor grown marijuana during 2020 was between \$340 and \$1100 a pound; high-quality, indoor grown \$1800–\$5000 a pound. This implies margins in excess of 200%. However, margin data provided by CSIMarket estimated gross margin in the legal marijuana industry to average 34.2% during 2018–2020.

The discrepancy can be attributed to the need to take account of indirect as well as direct costs. Indirect costs include taxes and licensing fees, fixed asset depreciation, and

TABLE 6 Leading marijuana companies traded on US exchanges

Symbol	Company	Market cap. ^a	Sales ^b	Net margin ^b	Notes
CGC	Canopy Growth Corporation	\$9700m.	\$478.0m.	(231%)	Canadian producer and supplier of medical cannabis products
CURLF	Curaleaf Holdings	\$8390m.	\$636.3m.	(9.5%)	Mass.-based producer and distributor of medical and recreational cannabis
GTBIF	Green Thumb Industries	\$5250m.	\$1213m.	(8.8%)	Chicago-based producer of branded cannabis products
CRLBF	Cresco Labs	\$3510m.	\$476.2m.	(17.4%)	Chicago-based producer of medical cannabis and related products
TCNNF	Trulieve Cannabis Corp.	\$3130m.	\$584.3m.	16.1%	Florida-based producer and distributor of medical cannabis products
CRON	Cronos Group Inc.	\$2780m.	\$50.2m.	254.2%	Canadian producer and supplier of cannabis and cannabis products under Peace, Naturals, Cove, Spinach, and Lord Jones brands
APHA	Aphria Inc.	\$2200m.	\$561.3m.	(22.8%)	Production and supply of medical cannabis. Based in Canada
GRWG	GrowGeneration Corp.	\$2120m.	\$157.0m.	3.2%	Denver-based supplier of growing systems
ACB	Aurora Cannabis Inc.	\$1770m.	\$271.0m.	(525%)	Production and distribution of cannabis products and indoor cultivation systems in Canada
TLRY	Tilray Inc.	\$1282m.	\$201.8m.	(226%)	Research, production, and distribution of medical cannabis, cannabinoids, and hemp-based food products. Based in Canada
HEXO:CA	HEXO Corp.	\$520.1m.	\$96.1m.	(511.0%)	Canadian producer and supplier of recreational and medical cannabis products
FFNTF	4Front Ventures Corp.	\$420.6m.	\$58.6m.	(17.7%)	Arizona-based producer and distributor of cannabis, CBD, and related products
SNDL	Sundial Growers Inc.	\$332.3m.	\$57.4m.	(223%)	Marijuana production and distribution in Canada
MMNFF	MedMen Enterprises Inc.	\$81.4m.	\$98.2m.	(122%)	Production and distribution of medical and recreational marijuana in California and four other states
ZENA:CA	Zenabis Global Inc.	\$34.8m.	\$79.1m.	165.6%	Canada-based producer and distributor of medical and recreational cannabis

Notes:^aOn December 18, 2020.^bMost recently available 12-month period.

overheads related to security, compliance, marketing, and administration. Moreover, as with all agricultural products, the crop losses are caused by disease, power outages, quality deficiencies, theft, and climatic adversity (including wildfires). The accounts of listed marijuana companies show dismal net margins—most of the leading marijuana companies traded on US exchanges earned losses during 2019 and 2020 (see Table 6).

Apart from licenses, the other costly inputs into the industry included growing facilities and skilled labor. As marijuana is legalized in an increasing number of states, so experienced personnel for cultivating and processing marijuana are in increasingly short supply. According to productiongrower.com, the average salary for cannabis master growers was \$125,000 in 2020.⁷

The greatest uncertainty in projecting future profitability relates to prices. In the wholesale market, prices are determined by supply and demand. Prices vary over times due to spikes in demand (demand peaks during summer and holidays) and seasonal variations in supply (supply from outdoor and greenhouse cultivation increases sharply during the fall). Figure 2 shows an index of wholesale prices between 2016 and 2020. For most of the period, there was a downward trend in prices as growth in supply outpaced growth in demand. Increasing demand during the COVID-19 pandemic caused an upturn in prices during 2020.

Within these averages, prices vary considerably in relation to quality and location. During the week ending December 18, 2020, Cannabis Benchmarks reported an average price of \$1530; 68% of the transactions it monitored were in the range \$1084–\$2548. In Massachusetts, Michigan, and Illinois, prices were more than double those in Washington and Colorado, while the price premium for high-quality, indoor-cultivated marijuana over outdoor-grown marijuana was between 60% and 200%.

Manufacturing

The recreational market has seen a wave of new businesses offering cannabis-infused foods, drinks, and personal care products. In addition to new start-ups, several established

FIGURE 2 Average spot wholesale price of marijuana in the United States (\$ per pound)



Source: Cannabis Benchmarks.

companies have added marijuana-infused products to their lineups, including a number of craft brewers such as Lagunitas Brewing Company, Coalition Brewing, and New Belgium Brewing Co. Statista estimated that in 2019, the top-five brands accounted for 80% of total sales of marijuana beverages. For edibles, the figure was 53%.

In the case of medical marijuana products, because of the longer development of the industry, more stringent purity and safety requirements, and greater commitment to research and development, larger enterprises dominate the market. Most of the biggest companies listed in Table 6 are engaged in the supply of medical marijuana products.

Retailing

Because of separate regulatory regimes and licensing procedures, medical and recreational dispensaries have tended to be separate, though increasingly, larger operators hold both medical and recreational licenses and, in some instances, operate combined facilities. Variations in regulations among states result in differences in size of establishments, operating practices, costs, and competitive conditions. In addition, some retailers are stand-alone, others backward integrated into cultivation.

The disparity between wholesale and retail prices of marijuana suggests attractive gross margins for marijuana dispensaries. However, the costs incurred in packaging, security, real estate, regulatory compliance, and marketing eat into these margins. Some indications of average revenues and costs are shown in Table 7.

Competition

All three major stages of the value chain—cultivation, manufacturing, and distribution—are fragmented. Most businesses are either owner proprietorships or partnerships. The number of businesses in each state depends upon the availability of licenses.

Like most agricultural products, marijuana is essentially a commodity product at the wholesale level, although there is extensive physical differentiation. Within the two

TABLE 7 Average revenues, costs, and profitability of licensed marijuana retailers

	Medical	Recreational ^a
Average outlet size (square feet)	2566	2100
Start-up cost (per outlet)	\$0.775m.	\$0.193m.
Annual operating cost (per outlet)	\$1.920m.	\$1.140m.
Revenue (per outlet)	\$3.000m.	\$1.800m.
Average profit margin ^b	15%	21%
Percentage of business that are profitable	65%	68%
Percentage of business that breakeven	29%	29%
Percentage of business that are loss-making	6%	3%

Notes:

^aIncludes combined medical and recreational stores.

^bAfter-tax, net margin.

Source: Based on Marijuana Business Daily, Marijuana Business Factbook 2017.

principal species of marijuana: *cannabis indica* and *cannabis sativa*, there are a huge number of different strains, each offering a different combination of characteristics and each comprising many different strains. Leafly.com (“The World’s Cannabis Information Resource”) lists and reviews over 3600 strains. Most of these strains have been created through selective breeding by growers and seed companies, and through naming these strains, brands are created.

Growers have had limited success in establishing their own brands—not least because of the inability to register trademarks for marijuana-based products with the US Patent and Trademark Office. At the retail level, differentiation has been greater—in addition to differentiation by geographical location, individual dispensaries can use quality and customer service to build customer loyalty.

Competition for commercially grown marijuana comes from several sources. Consumers, both medical and recreational, can legally grow their own in states where recreational marijuana is legal (in California, Colorado, and Washington, adults can cultivate up to six plants). In addition, there is illegal marijuana: both domestic and imported (mainly from Mexico and British Columbia). In 2019, it was estimated that illicit sales of marijuana in California totaled \$8.3 million, as compared to \$3.2 million of legal sales, and that there were 2800 unlicensed dispensaries, far outnumbering the 873 licensed dispensaries.⁸ The California Growers Association estimated that, in 2018, of the state’s estimated 68,150 cannabis growers, only 534 held licenses to cultivate cannabis.⁹ Lack of regulatory costs and taxes means that illegal producers have a huge cost advantage over legal producers. Moreover, the supply chains and distribution networks for illegal marijuana are well established.

The Future

Government policy will continue to exert a dominant influence over the development of the US marijuana industry in the coming years. Despite optimism that the Biden administration would favor a more permissive approach to marijuana at the federal level, most observers expect a less active approach to enforcement rather than outright legalization. While the medical benefits of CBD for a variety of health conditions are becoming increasingly recognized, medical opinion over the physical and psychological risks of recreational use of THC remains divided.

If the trend toward legality at the state level continues, but illegality persists at the federal level, it seems likely that the industry will continue its gradual consolidation. But as long as marijuana cannot be transported across state lines and access to banking and legal services remains limited, it is doubtful whether large, national players will emerge.

If legalized at the federal level, the development of the marijuana industry in Canada offers pointers to potential developments in the United States. Between 2017 and 2020, the Canadian marijuana industry consolidated at a rapid rate. The merger between Aphria Inc. and Tilray Inc. will result in two companies—Canopy Growth and Tilray-Aphria—emerging as the leading players in the global as well as the Canadian marijuana sector.

The Canadian marijuana industry also points to the potential involvement by tobacco and alcohol companies in the business. Tobacco and alcoholic beverages are the most developed industries offering intoxicating and addictive products that are harmful to health and highly regulated. The tobacco industry is also interesting because of its capacity to generate high profits, despite falling consumption, tight regulation, litigation, and heavy taxation. (Altria/Philip Morris, BAT, Japan Tobacco, and Imperial

Tobacco earned an average return on equity of 58% during 2015–2020.) Altria, the world's largest tobacco company, acquired 45% of Cronos in 2018, while US beer and wine giant, Constellation Brands, acquired a 10% stake in Canopy Growth, with a view to developing marijuana-infused beverages.¹⁰ Anheuser-Busch InBev and Molson Coors have both invested in Canadian marijuana producers.¹¹

In the medical field, big pharma interest in the potential offered by marijuana was triggered in 2018 by the British-based, GW Pharmaceuticals, obtaining FDA approval for its Epidiolex, CBD-based drug. Strategic alliances between pharmaceutical and medical marijuana companies include Novartis' agreement to distribute Tilray's cannabis-derived products, Johnson & Johnson's collaboration with Avicanna to research marijuana's medical potential, AbbVie's multiple research partnerships, and Teva Pharmaceutical's distribution agreements with InterCure and Syqe.¹²

Notes

1. <https://www.kearney.com/the-cannabis-opportunity/the-next-gold-rush-cannabis-and-consumer-products>, accessed December 13, 2020.
2. Colorado, Washington, Oregon, California, Nevada, Massachusetts, Maine, Alaska, Vermont, Michigan, Illinois, Montana, North Dakota, and Arizona.
3. National Institute on Drug Abuse, *Marijuana Research Report* (July 2020).
4. B. H. Han and J. J. Palamar, "Trends in Cannabis Use Among Older Adults in the United States, 2015–2018," *JAMA Internal Medicine* 180(4) (2020): 609–611.
5. *Cannabis Business Times* (June 2020): S4.
6. Ibid: S3.
7. <https://productiongrower.com/blogs/news/marijuana-grower-salaries-how-much-do-growers-make>, accessed December 20, 2020.
8. <https://spectrumnews1.com/ca/la-west/inside-the-issues/2019/11/13/audit-show-number-of-unlicensed-pot-shops-in-los-angeles-continues-to-increase>, accessed December 21, 2020.
9. <https://www.greenmarketreport.com/californias-cannabis-market-verge-crisis>, accessed December 21, 2020.
10. <https://www.forbes.com/sites/kriskrane/2018/12/19/cannabis-attracts-big-tobacco-alcohol-and-pharma-which-big-industries-will-join-next/>, accessed December 21, 2020.
11. <https://movendi.ngo/news/2020/09/19/addiction-industries-cannabis-attracts-big-tobacco-big-alcohol-and-pharma-investments/>, accessed December 21, 2020.
12. <https://www.fool.com/investing/2020/08/08/3-big-pharma-companies-with-surprising-cannabis-co/>, accessed December 21, 2020.